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Travels of the Cache Creek Terrane: a paleomagnetic, geobarometric and $^{40}Ar/^{39}Ar$ study of the Jurassic Fourth of July Batholith, Canadian Cordillera☆

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Abstract

The Middle Jurassic Fourth of July Batholith and cross-cutting mafic dikes have been studied geochronologically, geobarometrically and paleomagnetically to estimate subsequent tectonic motion of the Cache Creek Terrane (CCT) in the northern Canadian Cordillera. $^{40}Ar/^{39}Ar$ hornblende ages from a granodiorite phase are similar to U–Pb zircon ages and indicate rapid cooling of the batholith upon intrusion, suggesting that the magnetization age is coincident with the 173-Ma crystallization age. Argon ages of biotite from the granodiorite and two mafic dikes have similar ages of ~165 Ma, which dates cooling through ~280 °C.

Aluminum-in-hornblende geobarometry indicates differential uplift of the batholith across a north–south fault zone along Atlin Lake with >6 km more uplift on its eastern side. Also, the eastern side has been tilted downward to the south–southwest by ~9°.

Combined paleomagnetic data from 20 granitoid and 11 mafic dike sites yield an in situ paleopole at 55°W, 63°N ($d_p = 5°$, $d_m = 5°$) and a tilt-corrected paleopole at 81°W, 55°N ($d_p = 5°$, $d_m = 6°$). Compared to the 173-Ma reference pole for the North American craton, the tilt-corrected pole suggests a significant southward translation of $16.1 \pm 3.7°$ and a significant clockwise rotation of $107 \pm 7°$. The translation estimate is similar to the Jurassic Teslin Crossing pluton in the Stikine Terrane, however, the rotation estimate is very different. This could indicate that the Cache Creek Terrane was at a similar latitude of the Stikine Terrane, but the two were not yet amalgamated.
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Keywords: Paleomagnetism; Tectonic motions; Canadian Cordillera; Ar-age dating; Jurassic

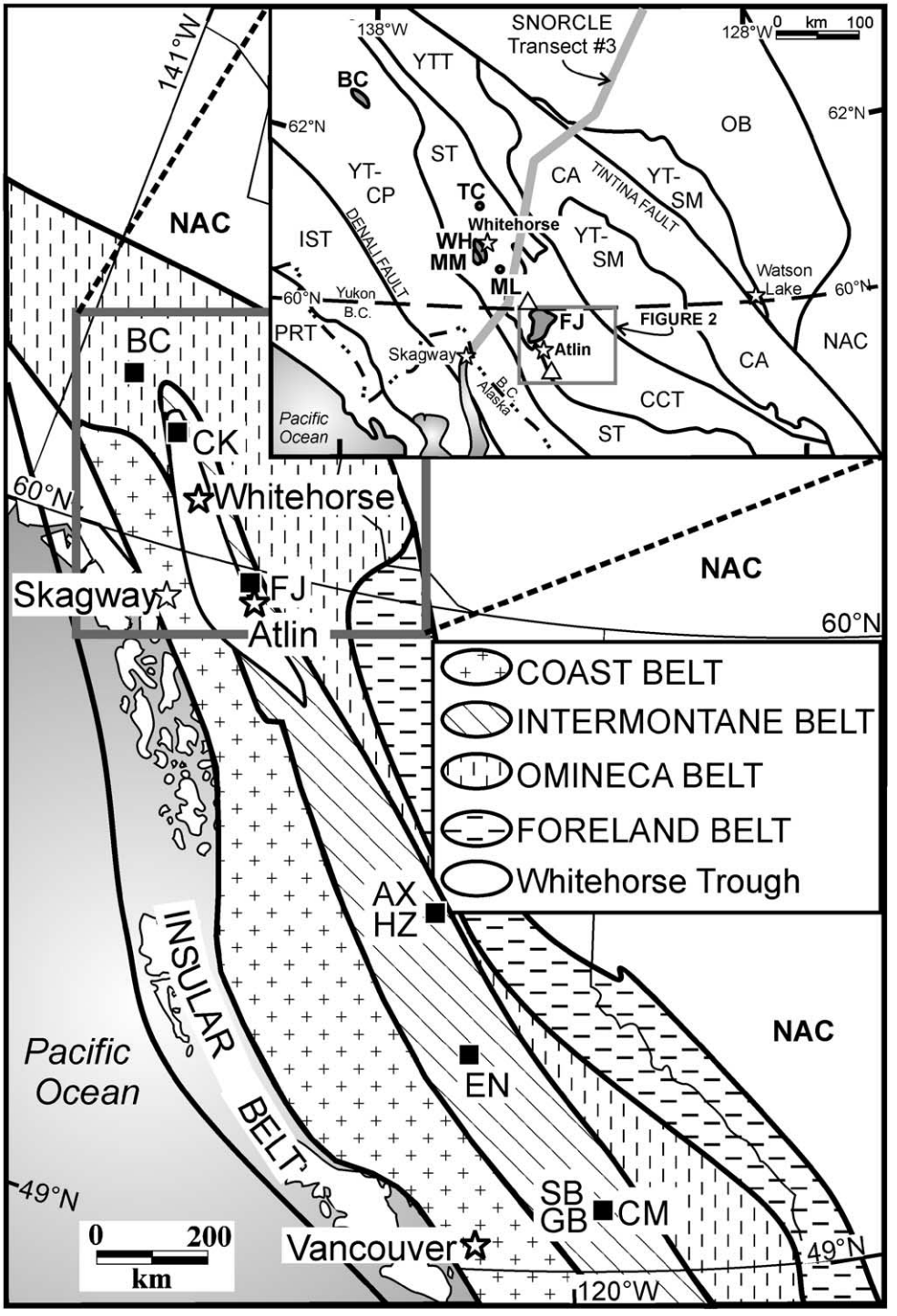
1. Introduction

Late Cretaceous and younger tectonic motions for terranes underlying the Intermontane Belt of the

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SNORCLE
Transect #3
138°W
128°W
0 km 100
62°N
60°N
YTT
BC
OB
ST
YT-
CP
CA
TINTINA FAULT
YT-
SM
TC
Whitehorse
WH
MM
ML
IST
DENALI FAULT
YT-
SM
Watson
Lake
Yukon
B.C.
PRT
FJ
Atlin
FIGURE 2
Skagway
B.C.
Alaska
CA
NAC
CCT
ST
Pacific
Ocean
141°W
NAC
BC
CK
Whitehorse
Skagway
FJ
Atlin
NAC
COAST BELT
INTERMONTANE BELT
OMINECA BELT
FORELAND BELT
Whitehorse Trough
AX
HZ
INSULAR
BELT
Pacific
Ocean
EN
NAC
SB
GB
CM
Vancouver
49°N
120°W
0
200
km

Canadian Cordillera are now relatively well documented (e.g., Gabrielse, 1985; Irving et al., 1995; Harris et al., 1997, 1999b; Wynne et al., 1998). However, there are few paleomagnetic results for rock units older than ~110 Ma and because of potential problems with age control and/or tectonic corrections, they have been viewed with skepticism by some authors. Nevertheless, as the paleomagnetic database increases, the larger terranes underlying the Intermontane Belt are beginning to show different tectonic motions prior to the mid to Late Jurassic amalgamation of the Stikine, Cache Creek and Quesnel terranes. This is the second paleomagnetic study of a Jurassic intrusion from the northern Canadian Cordillera. The first, from the Teslin Crossing Stock from the Stikine Terrane, suggested a trans-Pacific plate motion before amalgamation (Harris et al., 1999a).

The Middle Jurassic Fourth of July Batholith is located in northernmost British Columbia (Fig. 1). It was sampled to estimate the mid-Mesozoic tectonic motions of the northern Intermontane Belt as part of the LITHOPROBE–Slave Northern Cordilleran Lithospheric Evolution (SNORCLE) project. The batholith intrudes rocks of the northern Cache Creek Terrane (CCT), which lies inboard of the Stikine Terrane and outboard of the Yukon-Tanana–Slide Mountain and Quesnel terranes along the northwestern margin of the North American Craton (Fig. 1). The CCT is dominated by upper Paleozoic to lower Mesozoic oceanic rock sequences that contain Tethyan-type Permian faunal assemblages that are similar to coeval assemblages found in eastern Asia (Monger and Ross, 1971) but no where else in the Canadian Cordillera (Monger, 1977). The CCT sequences and assemblages are unlike the arc sequences of the neighbouring Stikine Terrane and the marine and nonmarine strata of the Yukon-Tanana–Slide Mountain and Quesnel terranes, suggesting that the CCT had a different pre-amalgamation history (Monger, 1977).

A previous paleomagnetic study in the northern CCT on volcanic, volcaniclastic and limestone strata near Atlin, British Columbia (Fig. 1; Cole et al., 1992) gave inconclusive results because of chemical remagnetization, lack of age control on the remagnetization, uncertain deformation events and other difficulties. The most conservative interpretation of Cole et al. (1992) postulated a post-folding Early Jurassic remagnetization when the Cache Creek and Stikine terranes were juxtaposed. This scenario placed the in situ paleopole near the Early Jurassic North American reference pole, suggesting little or no translation and minor counterclockwise rotation. They compared this interpretation to similar results from coeval rocks in the southern Stikine Terrane and suggested that the terranes of the Intermontane Belt had behaved as a coherent microplate since the Early Jurassic.

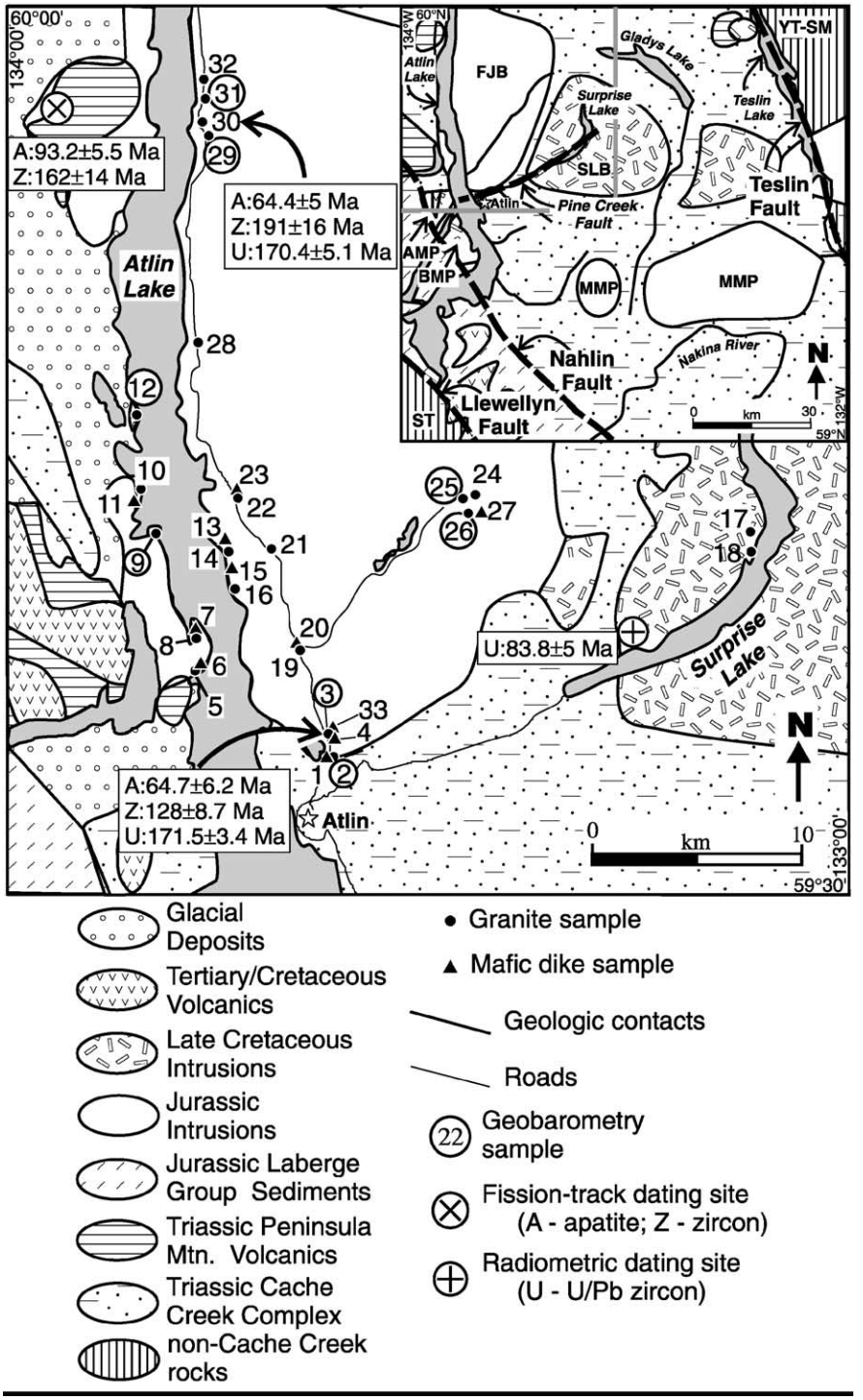
Here we report argon geochronometric, geothermobarometric and paleomagnetic data for the Fourth of July Batholith and cross-cutting dikes to estimate the tectonic motion of the Cache Creek Terrane since the Middle Jurassic.

2. Geology

2.1. Regional

The CCT is composed dominantly of the three lithostratigraphic assemblages of the Cache Creek Complex: (1) mafic volcanic rocks with minor ultramafic rocks; (2) chert and associated pelite; and (3) lenses of pure massive limestones in a matrix of chert or basalt. This sequence represents prolonged deposition in an ocean-floor environment from at least early Mississippian to Early Triassic time (Monger, 1977; Gabrielse and Yorath, 1991). In the study area (Fig. 2),

Fig. 1. The morphogeological belts of the Canadian Cordillera after Gabrielse et al. (1991b) and Harris (1998) and the tectonic subdivisions of the northern Cordillera after Harris (1998). Abbreviations are: AX—Axelgold Intrusions; BC—Big Creek Batholith; B.C.—British Columbia; CA—Cassiar Terrane; CCT—Cache Creek Terrane; CK—Carmacks Group sampling sites; CM—Copper Mountain Intrusions; EN—Endako Intrusions; FJ—Fourth of July Batholith; GB—Guichon Batholith; HZ—Hazelton Group sampling sites; IST—Insular Superterrane; ML—Mount Lorne Stock; MM—Mount McIntyre Stock; NAC—North American Craton; OB—Omineca Belt; PRT—Pacific Rim terranes; SB—Spences Bridge sampling sites; ST—Stikine Terrane; TC—Teslin Crossing Stock; WH—Whitehorse Batholith; YT–CP—Yukon-Tanana–Coast Plutonic Terrane; YT–SM—Yukon-Tanana–Slide Mountain Terrane; YTT—Yukon-Tanana Terrane. Triangles near Atlin, B.C. within the inset are paleomagnetic sampling areas of Cole et al. (1992).


60°00'
134°00'
A:93.2±5.5 Ma
Z:162±14 Ma
A:64.4±5 Ma
Z:191±16 Ma
U:170.4±5.1 Ma
Atlin Lake
U:83.8±5 Ma
Surprise Lake
A:64.7±6.2 Ma
Z:128±8.7 Ma
U:171.5±3.4 Ma
Atlin
0 km 10
133°00'
59°30'
FJB
SLB
AMP
BMP
MMP
MMP
Gladys Lake
Surprise Lake
Teslin Lake
YT-SM
Teslin Fault
Pine Creek Fault
Nahlin Fault
Llewellyn Fault
Nakina River
ST
0 km 30
59°N
132°W
Glacial Deposits
Tertiary/Cretaceous Volcanics
Late Cretaceous Intrusions
Jurassic Intrusions
Jurassic Laberge Group Sediments
Triassic Peninsula Mtn. Volcanics
Triassic Cache Creek Complex
non-Cache Creek rocks
Granite sample
Mafic dike sample
Geologic contacts
Roads
Geobarometry sample
Fission-track dating site (A - apatite; Z - zircon)
Radiometric dating site (U - U/Pb zircon)

the complex is overlain by Triassic to Jurassic mafic to intermediate volcanic and volcaniclastic rocks of the Peninsula Mountain Suite (Mihalynuk et al., 1992). On the west side of Atlin Lake, Late Cretaceous rhyolite, intermediate lapilli tuff and quartz-phyric ash-flow units of the Table Mountain Volcanic Complex unconformably overlie the Peninsula Mountain Suite. The area is intruded by numerous felsic plutonic bodies including: (1) the Middle Jurassic Fourth of July Batholith and Mount McMaster Stock; (2) the Late Cretaceous Surprise Lake Batholith and Atlin Mountain Pluton; and (3) the Paleocene Birch Mountain Pluton (Fig. 2; Aitken, 1959; Mihalynuk et al., 1992). West of Surprise Lake, Pleistocene olivine basalt and scoria are found unconformably on the Cache Creek Complex and the Surprise Lake Batholith (Aitken, 1959).

The Atlin area of the CCT is fault-bounded with the Llewellyn and Nahlin Fault zones to the west and the Teslin Fault Zone to the east (Fig. 2, inset). The Llewellyn Fault Zone is a dextral west-side-up transcurrent fault that most likely first formed in the Triassic and has since been reactivated to cut rocks at least as young as the mid-Cretaceous (Mihalynuk and Mountjoy, 1990). South of the study area, the Llewellyn Fault appears to truncate the northwest-trending King Salmon Thrust Fault, which was the locus of southwest-directed thrusting of the Cache Creek Terrane over the Stikine Terrane (Gabrielse, 1985; Gabrielse et al., 1991a). The Nahlin Fault Zone abuts upper Paleozoic CCT rocks to the north against dominantly Mesozoic island arc-related rocks of the Stikine Terrane to the south (Fig. 2; Aitken, 1959). This zone parallels both the Llewellyn and King Salmon Thrust faults and thus is likely related to the amalgamation of the two terranes. The Teslin Fault Zone separates the CCT to the west from the Yukon-Tanana–Slide Mountain Terrane and Quesnel Terrane to the east. This fault zone is part of a younger system of north–south-trending dextral strike-slip faults that have a combined displacement of about 200 km (Gabrielse et al., 1991a).

2.2. Fourth of July Batholith

The Fourth of July Batholith has an outcrop area of ~ 650 km^2 (Fig. 2). It is composed of pink granite, granodiorite, tonalite and monzonite with subordinate biotite- and hornblende-rich phases and potassium feldspar megacrystic to equigranular phases (Mihalynuk et al., 1991). Micropetrographic observations show that the granitoid sites are dominantly comprised of plagioclase (An_8 to An_{46}, mean An_{22}, from microprobe analyses), potassium feldspar, quartz, biotite, hornblende and minor amounts of clinopyroxene. Trace minerals include apatite, titanite and opaque minerals, with the latter being mainly magnetite. There is minor alteration throughout the batholith which includes: biotite and hornblende to chlorite, feldspars to sericitic mica, and plagioclase to calcite and clay minerals. The batholith's intrusive contact is mostly covered, however, it is sharp and irregular where observed. The bedding of the country rocks is generally steeply dipping and parallel to the observed or inferred contacts (Aitken, 1959). Deformation of the host rocks is thought to have occurred during terrane amalgamation and obduction in Late Toarcian to Bajocian time (i.e., 183 ± 7 Ma; Cordey et al., 1987; Mihalynuk et al., 1992).

A combined U–Pb zircon isochron age of 172 ± 3 Ma is thought to date crystallization of the Fourth of July Batholith most accurately (Table 1, Fig. 2), although the measured ages range from 167 to 174 Ma (Mihalynuk et al., 1992). Zircon and apatite fission-track dating by Donelick and Dickie (1991) on nine samples from three different locations range from 215 to 128 Ma and from 108 to 47 Ma, respectively (Table 1, Fig. 2).

2.3. Mafic dikes

The southern part of the Fourth of July Batholith is cut by mafic dikes that are thought to be late stage

Fig. 2. Geology around the Fourth of July Batholith after Aitken (1959) with sampling sites and the distribution of igneous rock bodies and fault systems in the study area. Abbreviations are: AMP—Atlin Mountain Pluton; BMP—Birch Mountain Pluton; FJB—Fourth of July Batholith; MMP—Mount McMaster Pluton; SLB—Surprise Lake Batholith; ST—Stikine Terrane; YT–SM—Yukon-Tanana–Slide Mountain Terrane. Ages are averages from Table 1: A—apatite fission track; Z—zircon fission track; U—U–Pb zircon (Donelick and Dickie, 1991; Mihalynuk et al., 1992).

Table 1
Fission-track and best-estimate radiometric ages for the Fourth of July Batholith

Location	Ap FT	Zir FT	U–Pb
South	64.7 ± 6.2	128 ± 8.7	171.5 ± 3.4
Northwest	108.0 ± 5.8	193 ± 18	
	88.0 ± 5.2	158 ± 12	
	83.8 ± 5.5	136 ± 11	
Northeast	FAIL	132 ± 10	170.4 ± 5.1
	47.0 ± 4.0	210 ± 19	
	68.0 ± 4.3	194 ± 16	
	75.7 ± 5.2	204 ± 17	
	79.9 ± 5.2	215 ± 20	

Locations are plotted in Fig. 2. All ages in Ma.
Ap FT—apatite fission-track ages, Zir FT—zircon fission-track ages (Donelick and Dickie, 1991); U–Pb—uranium–lead zircon ages (Mihalynuk et al., 1992). Errors are one standard deviation for the fission-track ages and 2σ (sigma) for the U–Pb ages.

intrusions of the batholith because they have similar compositions to the more mafic portions of it (Mihalynuk et al., 1991; Mihalynuk and Smith, 1992). The dikes that were collected are 0.3 to 10 m wide and have an average trend of 293 ± 24° and dip 75 ± 16° NNE. These dikes, dominantly lamprophyric in composition, are massive with fine-grained chilled margins and medium-grained centers (Aitken, 1959; Mihalynuk and Smith, 1992). Micropetrographic observations show that the major minerals are plagioclase, clinopyroxene and biotite and the minor minerals include quartz, hornblende and opaques. Minor alteration includes plagioclase to calcite and clay minerals, clinopyroxene to chlorite, and hornblende is pseudomorphic after clinopyroxene.

2.4. Surprise Lake Batholith

The Surprise Lake Batholith is a polyphase body of texturally diverse alaskite, granite, and minor quartz monzonite that intrudes the eastern margin of the Fourth of July Batholith as well as the supracrustal rocks of the CCT (Fig. 2). Mineralogically, the Surprise Lake Batholith includes smoky quartz, chalky albite, potassium feldspar and biotite (Bloodgood et al., 1989) with accessory minerals that reflect U, W, Sn and F enrichment suggesting that it is chemically evolved (Ballantyne and Littlejohn, 1982). The weighted mean of the $^{207}Pb/^{206}Pb$ ages for zircon fractions yielded a crystallization age of 84 ± 5 Ma (Mihalynuk et al., 1992). Also reported are an average K–Ar biotite age of 70.6 ± 3.8 Ma (Christopher and Pinsent, 1982) and a whole rock Rb–Sr age of 72.2 ± 0.8 Ma (Mihalynuk et al., 1992).

3. $^{40}Ar–^{39}Ar$ geochronology

3.1. Methods

Hand-picked biotite and hornblende crystals were packaged within 3 × 3 mm aluminum packets and irradiated in the research reactor of McMaster University for 8 h in a fast neutron fluence of about 4×10^{13} neutrons/cm^2/s, with a resulting J-factor of 0.002. Standard FCT-SAN, with an apparent age of 28.0 ± 0.1 Ma (Renne et al., 1994), was incorporated within the packets of all samples resulting in over 40 individual measurements of the flux monitor. Best-fit regression yielded an estimate of flux gradient for each individual sample. Upon return from the reactor, the samples were split into multiple–single or multi-grain aliquots, loaded into individual 1.5-mm diameter holes in a copper planchet, heated and analysed separately. Heating of samples was done with a Merchantek® MIR-10 10W CO_2 laser that is equipped with 2 × 2-mm flat-field lens, obviating the need for beam panning. Data were acquired with a VG3600 gas source mass spectrometer at the Geological Survey of Canada laboratory in Ottawa. Data collection protocols followed those detailed in Villeneuve and MacIntyre (1997) and data reduction procedures for individual aliquots followed Roddick (1988). Initially, each aliquot of a sample was treated as a separate step-heated analysis, with data reduction procedures for each aliquot following Roddick (1988). Data from individual aliquots are plotted on Ar gas release spectra in Fig. 3. These plots differ from conventional $^{40}Ar/^{39}Ar$ spectra in that data from individual aliquots are displayed in adjacent, alternately shaded regions of gas release spectra. Upon ascertaining reproducibility of individual spectra and plateau regions between aliquots, data were combined by integrating plateau portions. In most cases, the two aliquots gave identical ages within error and the data were combined and treated as a single step-heated analysis for data reduction. Suitable heating steps, i.e., those without indication of ^{40}Ar-loss, were

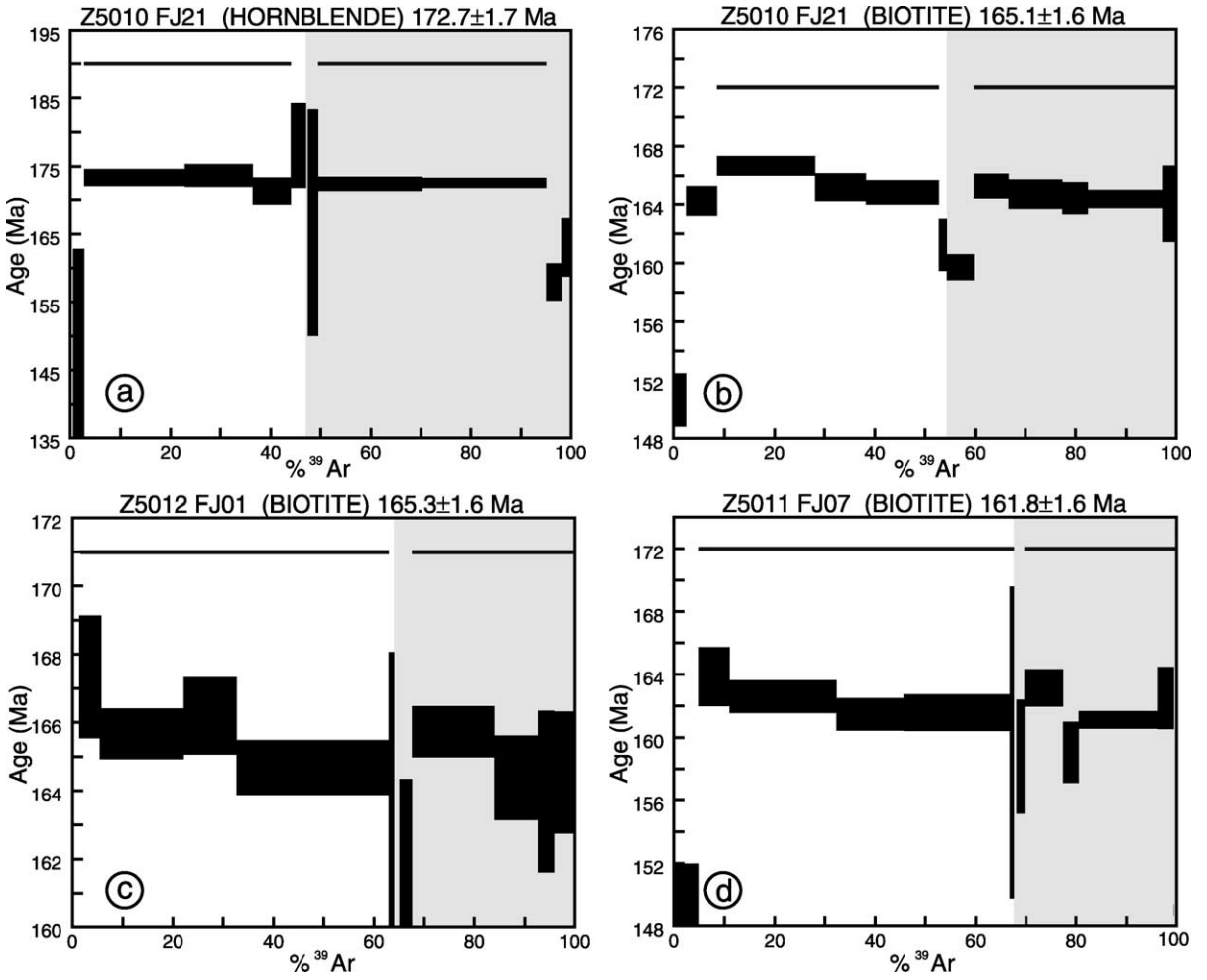


Fig. 3. Argon gas release spectra for granitoids (a, b) and mafic dikes (c, d) in the Fourth of July Batholith. White regions represent aliquot A and grey regions represent aliquot B (see text for details).

selected and regressed utilizing a modified York (1969) technique, anchored at $^{40}Ar/^{36}Ar = 295.5$. For reasons outlined in Roddick (1988), error in the J-factor measurement of $\pm 0.5\%$ (1σ) is only included in the final age determination. Uncertainty on apparent age of the monitor is not propagated into the final age. All errors herein are presented at the 2σ level of uncertainty unless otherwise noted.

3.2. Results

$^{40}Ar/^{39}Ar$ spectra were determined for three samples from the Fourth of July Batholith and associated dikes. A sample of coarse-grained biotite–hornblende granodiorite (site FJ21) is from the east side of Atlin Lake (Fig. 2). Hornblende from it gives flat, reproducible plateaux on two aliquots, with only minor evidence of Ar-loss in the lowest temperature steps of aliquot A. Ar-loss in the highest temperature steps of aliquot B may be the result of degassing of inclusions or small previously unheated grains that had adhered to the wall of the sample hole. The majority of the gas that forms the plateau segments results in an age of 172.7 ± 1.7 Ma (Fig. 3a). Biotite from the same sample also yields relatively flat reproducible plateau segments with clear evidence of

Ar-loss in the earliest heating steps. The resulting age of 165.1 ± 1.6 Ma (Fig. 3b), however, is clearly distinct from the hornblende age, suggesting final passage through the ca. 280 °C closure temperature of biotite at that time.

A biotite sample was separated from a lamprophyre dike that cuts the southern end of the batholith on the east side of Atlin Lake (site FJ01; Fig. 2). It displays similar systematics and age to the biotite sample from the granodiorite, with reproducible plateau segments on two aliquots, yielding an age of 165.3 ± 1.6 Ma (Fig. 3c). Biotite from a second lamprophyre dike located on the west side of Atlin Lake was also analysed (site FJ07; Fig. 2). In this case, the two aliquots displayed minor Ar-loss in the earliest heating steps yet still give similar plateau ages, although aliquot B appears to be slightly "noisier" than aliquot A. The derived age of 161.8 ± 1.6 Ma (Fig. 3d) is measurably younger than the biotite age from the previous two samples.

4. Geothermobarometry

Using the aluminum-in-hornblende geobarometer and the associated hornblende–plagioclase geothermometer, amphibole and plagioclase crystals from eight sites were analysed to estimate the amount of post-crystallization tilting of the Fourth of July Batholith. The theory and procedures have been described in detail elsewhere (e.g., Hammarstrom and Zen, 1986; Blundy and Holland, 1990; Ague and Brandon, 1992; Holland and Blundy, 1994; Anderson and Smith, 1995; Harris et al., 1997, 1999b). The operating conditions on the JEOL JXA 8600 electron microprobe at the University of Western Ontario were set at: a probe current of 10 nA; an accelerating voltage of 15 kV; a focused beam of 2 μm for amphibole and 5 μm for plagioclase; and a 20-s accumulation time.

Representative compositions of amphibole crystal edges from the Fourth of July Batholith are given in Table 2. The amphibole structural formulae are calculated on the basis of 13-cations exclusive of Ca, Na and K (Cosca et al., 1991; Harris et al., 1997) and the plagioclase structural formulae are calculated on the basis of 32 oxygens. The amphiboles show a small compositional range in Si cations versus a wide range in the Mg-ratio and they are classified as magnesiohornblendes and ferrohornblendes (Table 2, Fig. 4a). $Fe^{3+}/(Fe^{3+}+Fe^{2+})$ ranges from 0.12 to 0.57 and $(Fe^{2+}+Fe^{3+})/(Fe^{2+}+Fe^{3+}+Mg)$ ranges from 0.33 to 0.62 (Table 2), thus satisfying oxygen fugacity restrictions as described by Anderson and Smith (1995), i.e., ~ 25 and 0.40–0.65, respectively. Except for site FJ03, the standard deviations of total aluminum are less than 9% of the average (Table 3), indicating a high internal precision for the six or more analyses per site.

Table 2
Representative amphibole chemical analyses and mineral formula compositions from the Fourth of July Batholith

Sample site	FJ02	FJ09	FJ25	FJ29
Sample	02041c	09052c	25051b	29032b
SiO_2	46.61	48.56	45.42	44.75
TiO_2	1.09	0.94	1.22	1.31
Al_2O_3	5.66	4.87	6.73	8.28
Fe_2O_3*	4.44	7.20	5.17	5.16
Cr_2O_3	0.00	0.02	0.03	0.00
FeO	16.40	7.23	18.50	12.55
MnO	0.69	0.83	0.74	0.44
MgO	9.77	15.10	7.98	11.40
CaO	11.40	11.39	11.37	11.84
Na_2O	1.34	1.67	1.40	1.31
K_2O	0.69	0.50	0.76	0.93
H_2O*	1.76	1.73	1.81	1.89
F	0.47	0.70	0.33	0.17
Cl	0.06	0.04	0.10	0.18
F, Cl=O	0.21	0.30	0.16	0.11
Total	100.17	100.48	101.40	100.09
Si	7.002	7.013	6.838	6.660
$^{[4]}Al$	0.998	0.829	1.162	1.340
$^{[6]}Al$	0.005	0.000	0.032	0.112
Fe^{3+}	0.502	0.783	0.586	0.577
Fe^{2+}	2.060	0.860	2.330	1.560
Ti	0.123	0.102	0.138	0.147
Mg	2.187	3.250	1.790	2.528
Mn	0.088	0.102	0.094	0.055
Cr	0.000	0.002	0.004	0.000
Ca	1.835	1.762	1.834	1.888
$^{[M4]}Na$	0.165	0.126	0.166	0.112
$^{[A]}Na$	0.225	0.341	0.243	0.266
K	0.132	0.092	0.146	0.177
OH	1.761	1.671	1.817	1.875
F	0.223	0.320	0.157	0.080
Cl	0.015	0.010	0.026	0.045
Fe^{3+}#	0.196	0.477	0.201	0.270
Fe#	0.539	0.336	0.620	0.458

Notes: Fe_2O_3* and H_2O* are back-calculated from the structural formula. $Fe^{3+}\# = Fe^{3+}/(Fe^{3+}+Fe^{2+})$; $Fe\# = Fe_{tot}/(Fe_{tot}+Mg)$.

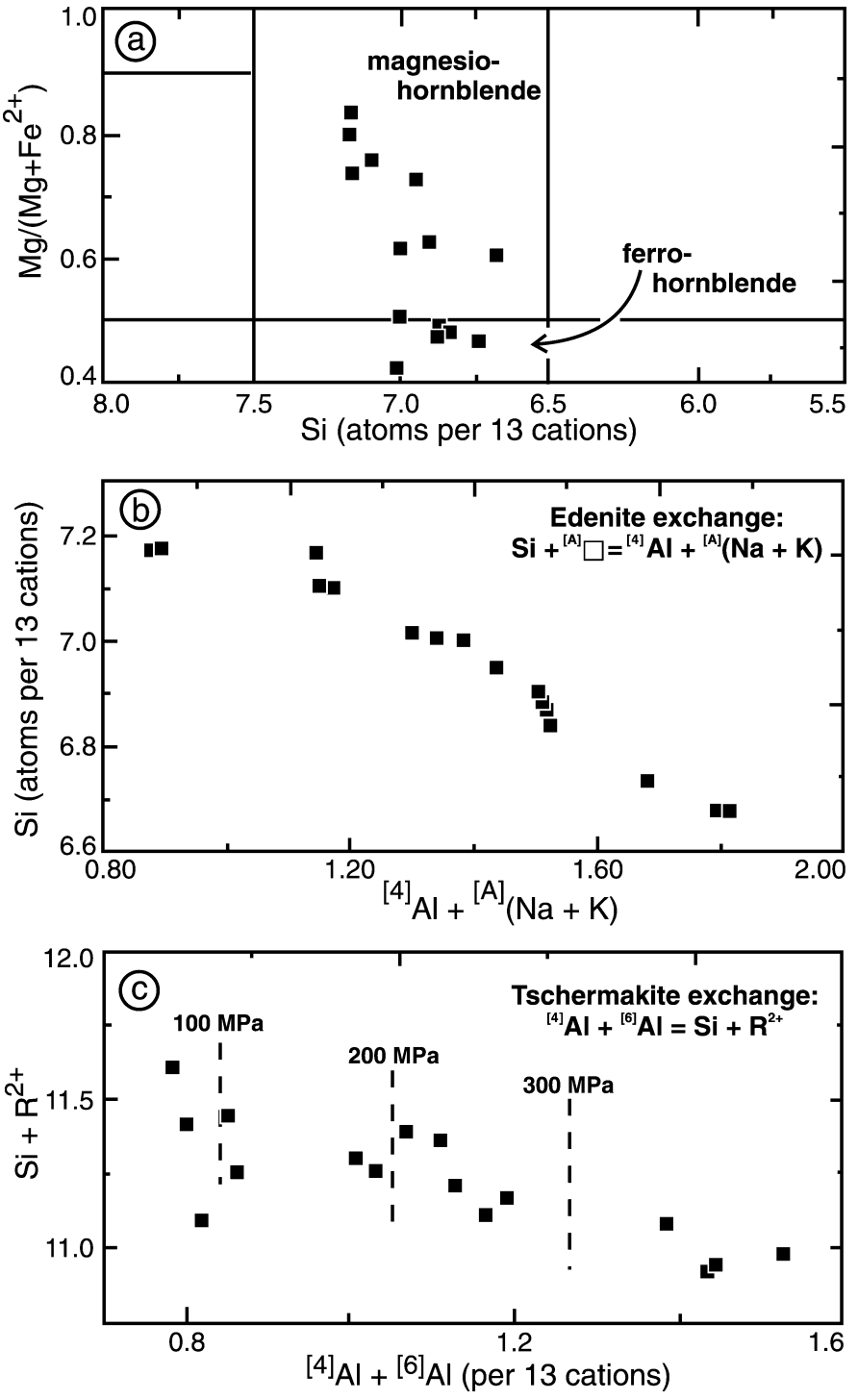


Fig. 4. Hornblende compositions of the Fourth of July Batholith. (a) Simplified calcic-amphibole classification plot after Leake et al. (1997). (b) and (c) are amphibole exchange reactions. Pressures shown in (c) are based on the Schmidt (1992) calibration.

Table 3
Site-averaged compositions of co-existing amphibole and plagioclase edges and raw thermobarometric data for the Fourth of July Batholith

Site	Plag (Ab#)	Amph (Si)	Amph (Al^{tot})	Std Dev.	P(Sch) (MPa)	T(B&H) (°C)	P(A&S) (MPa)	Depth (km)
FJ02	0.82	6.95	1.07	0.10	208	681	204	7.3
FJ03	0.89	7.06	0.90	0.13	128	657	136	4.9
FJ09	0.91	7.14	0.79	0.04	76	645	86	3.1
FJ12	0.55	7.17	0.84	0.03	97	711	70	2.5
FJ25	0.84	6.89	1.19	0.05	267	681	263	9.5
FJ26	0.86	6.95	1.10	0.11	222	672	224	8.1
FJ29	0.68	6.68	1.44	0.02	384	744	292	10.5
FJ31	0.62	6.79	1.45	0.11	389	737	309	11.1

Plag Ab# is the atomic ratio (Na/(Na + Ca)) in plagioclase. Amph (Si) and Amph (Al^{tot}) are the cation proportions of silica and total aluminum in the amphibole structure, respectively. Std Dev. is the standard deviation of the Amph (Al^{tot}) average. P(Sch) is from Schmidt (1992), T(B&H) is from Blundy and Holland (1990) and P(A&S) is from Anderson and Smith (1995).

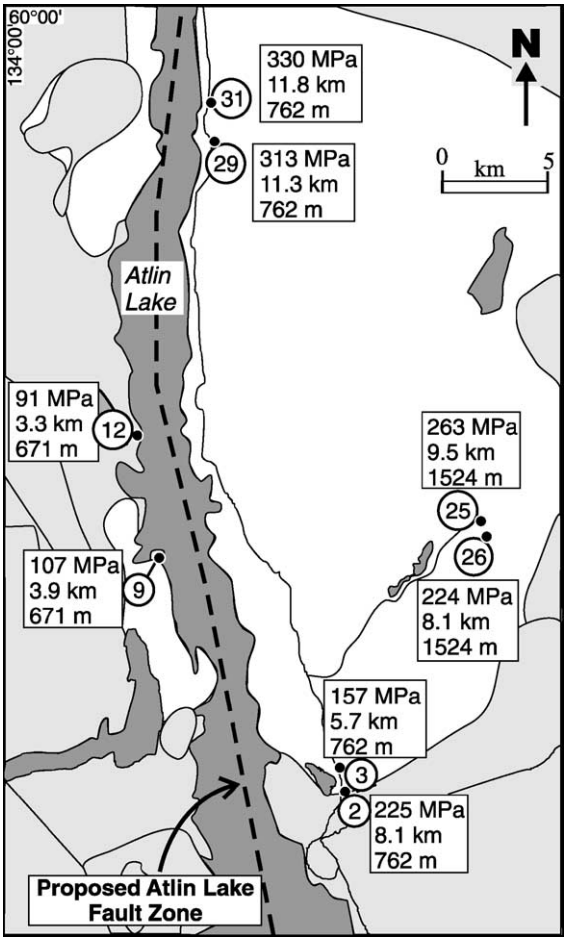


Fig. 5. Topographically corrected pressure, depth and elevation data for the Fourth of July Batholith.

Both the temperature-sensitive edenite exchange reaction (Fig. 4b) and the pressure-sensitive tschermakite-forming reaction (Fig. 4c) show linear trends and generally reflect the low pressures of crystallization. Preliminary mean pressures range from 76 to 389 MPa (Table 3, Fig. 4c) and mean temperatures range from 645 to 744 °C (Table 3). The temperature-correction on the geobarometer of Anderson and Smith (1995) yields a range in mean pressure from 70 to 309 MPa (Table 3). The geothermometer carries a ± 50 °C error and the geobarometer carries a ± 60 MPa error. There is a geographical distribution to the pressure estimates with the lowest two pressure values lying to the west of Atlin Lake, the highest two pressure values lying to the north, and a gradual decrease southward towards Atlin (Fig. 5). Intrusion depths were estimated from these pressure values using an assumed crustal density of 2.8 g/cm^3 and a geobarometric gradient of 3.6×10^{-2} km/MPa (Table 3, Fig. 5).

5. Paleomagnetism

5.1. Methods

Thirty-three sites were sampled north of Atlin (Fig. 2) with at least five cores drilled per site and oriented in situ, generally using a solar compass. Of the 33 sites, 20 were in the granitoids of the Fourth of July Batholith; 11 were sampled in the mafic dikes with two of these sites, FJ06 and FJ33, providing paleomagnetic contact tests between the dikes and the granitoid host; and two sites were sampled in the

Surprise Lake Batholith (Fig. 2) for another paleomagnetic contact test.

All paleomagnetic measurements were done at the University of Windsor using an automated CTF DRM-420 cryogenic magnetometer, a Sapphire Instruments SI-4 alternating field demagnetizer, a Sapphire Instruments SI-5 spinner magnetometer, a Sapphire Instruments SI-6 pulse magnetizer, and a Magnetic Measurements MMTD-1 thermal demagnetizer. All the instruments are situated within a magnetically shielded room that has an ambient field of ~ 0.2% of the Earth's magnetic field intensity.

The natural remanent magnetization (NRM) was measured for each of the 360 specimens. The median NRM intensity for the granitoids and the dikes are 2×10^{-1} and 1×10^{0} A/m, respectively. Three specimens with average NRM intensities and directions were selected as pilot specimens for each site. Two of the specimens were subjected to alternating field (AF) demagnetization in 10 steps from 5 to 130 mT and the third was subjected to thermal demagnetization in 12 steps from 200 to 600 °C. Most sites showed that the specimens had magnetic coercivities and thermal unblocking temperatures typical of magnetite but a few sites had specimens with pyrrhotite unblocking temperatures. Since both demagnetization methods gave similar results for a given site, all remaining specimens were AF demagnetized using at least five steps from 20 to 100 ± 20 mT because it is the more efficient method. Specimens from six of the sites were subsequently demagnetized thermally, with three to five steps in both the pyrrhotite (260–320 °C) and magnetite (500–570 °C) unblocking temperature ranges. Saturation isothermal remanent magnetization (SIRM) testing was done on 20 of the AF pilot specimens that represented the range of rock types and magnetic minerals. Each specimen was pulse magnetized in 11 steps from 10 to 900 mT and then AF demagnetized in seven steps from 10 to 140 mT. Specimen ChRM directions were obtained using the end-point method of Kirschvink (1980) and the remagnetization circle method of Halls (1978). Site and unit mean directions were calculated using Fisher (1953) statistics.

5.2. Results for the Fourth of July Batholith

Thermal demagnetization on the granitoid specimens shows a range of unblocking temperatures that are indicative of: (1) pure or fine-grained magnetite (specimens FJ05, FJ19); (2) coarse-grained or titaniferous magnetite (FJ02, FJ16); and, (3) pyrrhotite (FJ25, FJ28) (Fig. 6a). SIRM acquisition and decay

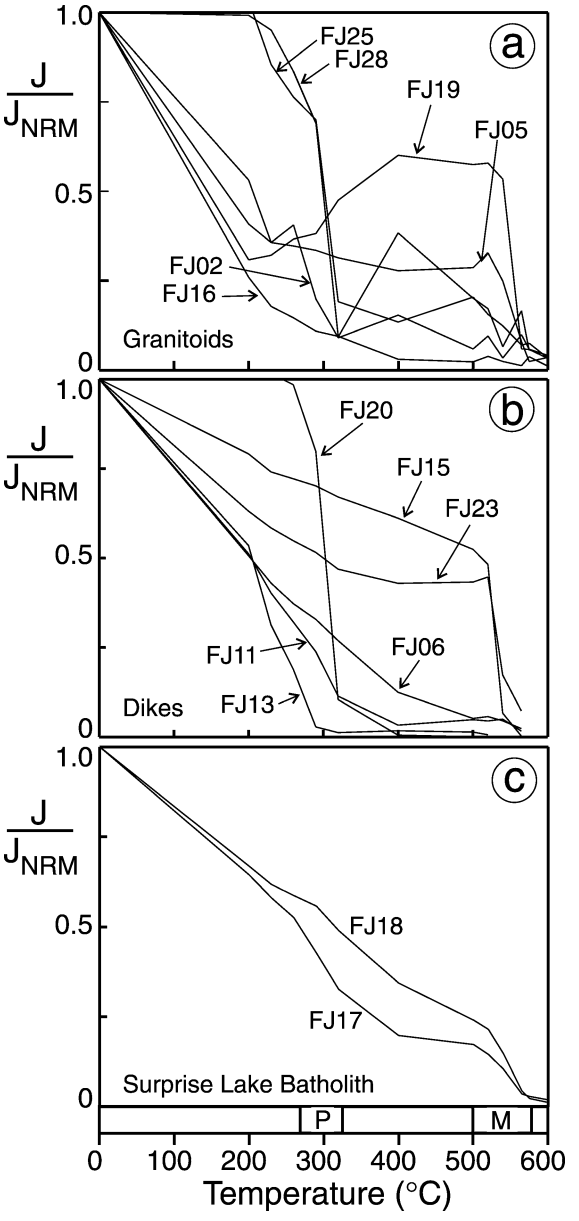


Fig. 6. Thermal decay curves for (a) the Fourth of July granitoids, (b) the mafic dikes and (c) the Surprise Lake granitoids. Expected unblocking temperatures for pyrrhotite and magnetite are shown at the bottom by the P and M, respectively.

curves also show the same range of magnetic minerals from multidomain to pseudosingle domain magnetite (FJ09, FJ22, FJ29) (Fig. 7a) and multidomain to single domain pyrrhotite (FJ02, FJ03, FJ25) (Fig. 7b). Orthogonal step demagnetization plots (Zijderveld, 1967) for the granitoids show that a viscous remanent magnetization (VRM) is present but that it is removed by 20 mT (Fig. 8a,b) or 200 °C and that two

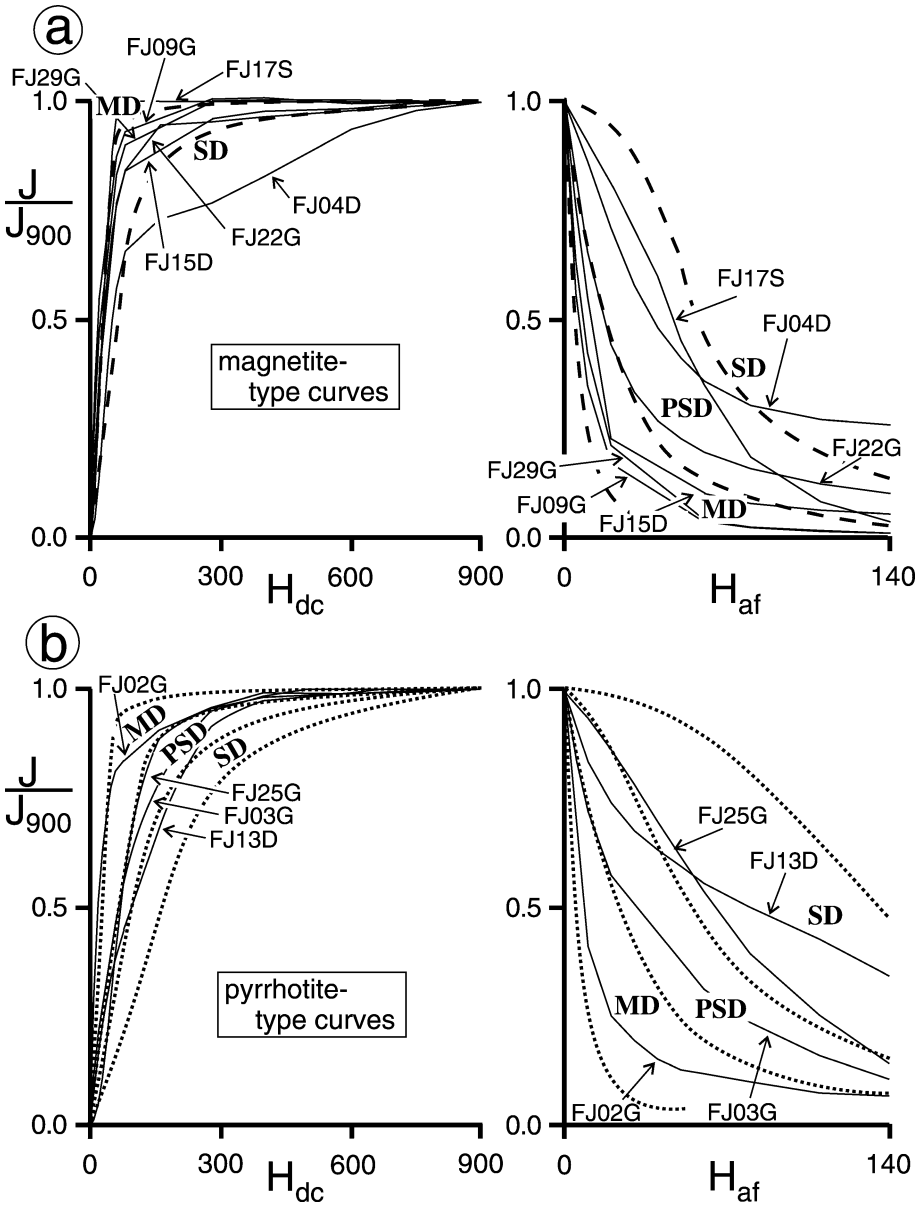


Fig. 7. SIRM acquisition and decay curves. The H_{dc} and H_{af} axes are measured in mT and the *y*-axis is the ratio of measured intensity versus the saturation intensity at 900 mT. The reference curves in (a) are for magnetite and in (b) are for pyrrhotite. Abbreviations are SD—single domain, PSD—pseudosingle domain, MD—multidomain. Specimen labels indicate the curve is for (G) Fourth of July granitoid; (D) mafic dike; or (S) Surprise Lake granitoid.

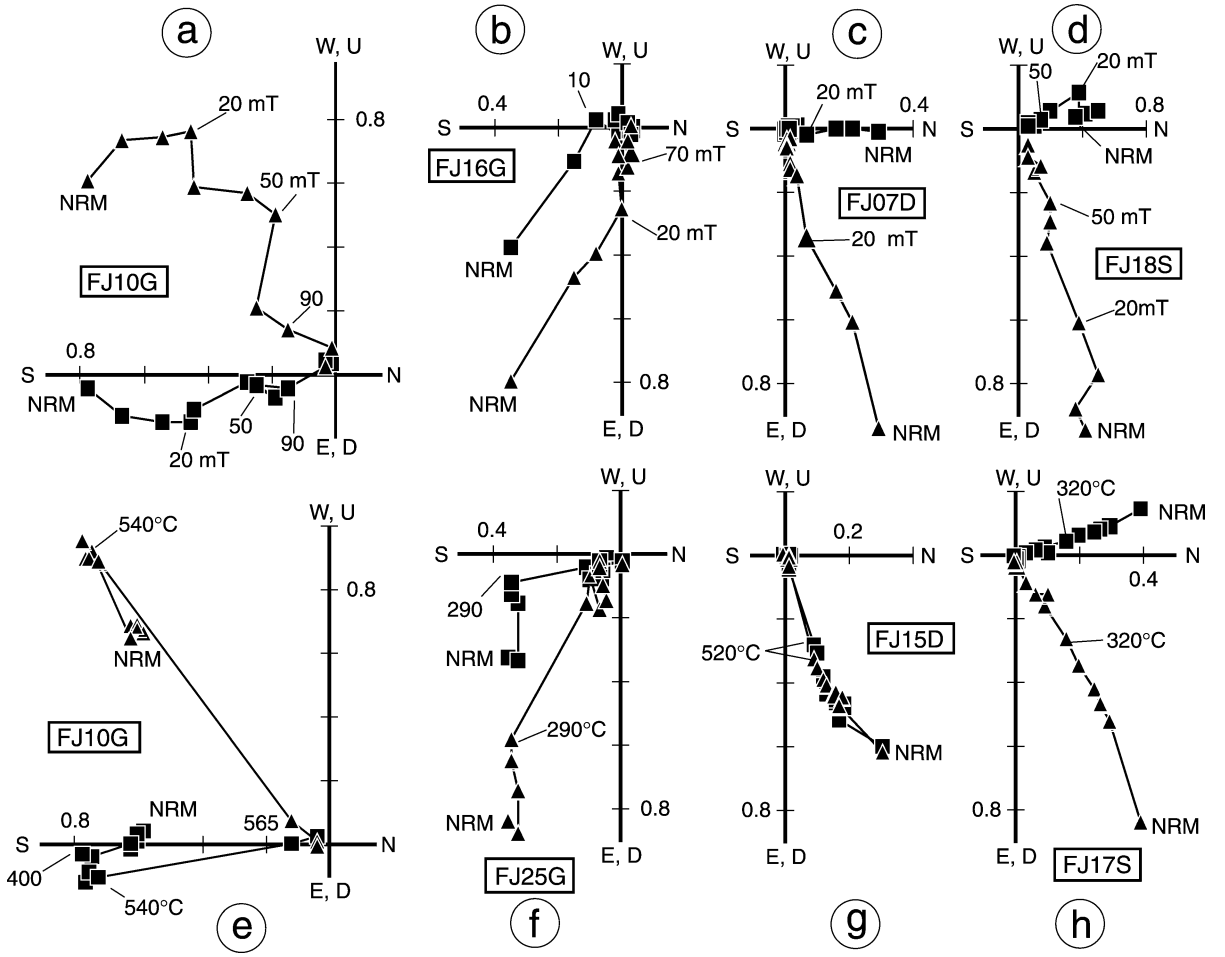


Fig. 8. Representative orthogonal step demagnetization plots (Zijderveld, 1967). Squares are projected in the north–east–south–west (N–E–S–W) plane and the triangles are projected in the north–down–south–up (N–D–S–U) plane. Specimen labels as defined in Fig. 7.

components may be present, one with a low unblocking and one with a high unblocking temperature range (Fig. 8e,f).

All 20 granitoid sites gave acceptable mean ChRM directions with radii of cones of 95% confidence (α_{95}) of $<13°$ (Table 4). The means showed a bimodal distribution with 14 sites directed northeast and steeply down and the other 6 sites antiparallel (Fig. 9a). The *F*-statistic of McFadden and Lowes (1981) showed that the two clusters, excluding polarity, can be drawn from one population at the 95% confidence level (i.e., $F_{calc} = 1.81 < F_{tab} = 3.23$). Therefore, combining all 20 site mean directions, after reversing the up-directed inclinations, yields a unit mean direction for the Fourth of July Batholith at declination (D) = 48.2°, inclination (I) = 72.3°, $\alpha_{95} = 3.6°$, and precision parameter (k) = 85 (Table 4). Bimodal mean ChRM directions are expected because the Earth's magnetic field changed polarity frequently during the Middle Jurassic (Opdyke and Channell, 1996).

5.3. Results for the mafic dikes

Specimens from the dikes show a wide range of thermal unblocking temperatures that indicate the presence of pyrrhotite (FJ13, FJ20), impure or

Table 4
Site-averaged in situ paleomagnetic data for the Fourth of July Batholith collection

Site	*N*, *e*, *r*	*D* (°)	*I* (°)	α_{95} (°)	*k*
FJ01d[a]	11, 6, 4	19.8	57.8	7.4	43
FJ02	7, 4, 3	241.9	−73.4	10.5	34
FJ03	9, 5, 3	60.7	71.1	11.3	25
FJ04d	11, 5, 4	68.5	62.0	5.7	82
FJ05	10, 3, 6	216.2	−63.6	6.5	63
FJ06d	10, 6, 3	42.4	78.8	8.1	41
FJ07d	14, 4, 5	59.0	72.2	6.7	60
FJ08	15, 8, 7	40.9	66.5	5.9	43
FJ09	11, 6, 5	53.0	83.5	6.6	50
FJ10	14, 7, 7	210.4	−75.4	6.0	45
FJ11d	18, 15, 1	223.7	−72.7	2.7	195
FJ12	11, 8, 3	35.6	70.9	8.3	32
FJ13d	13, 4, 6	33.2	59.7	8.4	34
FJ14	9, 4, 4	66.5	68.6	9.7	34
FJ15d	16, 7, 9	70.5	61.4	5.2	51
FJ16	10, 4, 5	53.4	76.8	7.8	45
FJ17s[a]	7, 4, 2	16.9	77.1	4.2	258
FJ18s	8, 4, 4	355.0	67.9	5.8	93
FJ19	10, 4, 5	234.8	−56.4	4.1	158
FJ20d	10, 10, 0	40.1	70.0	2.8	301
FJ21	12, 3, 8	70.7	66.7	4.8	90
FJ22	7, 6, 1	357.7	79.9	7.2	71
FJ23d	11, 11, 0	62.9	61.6	8.0	33
FJ24	8, 6, 2	215.8	−74.5	11.8	23
FJ25	8, 5, 1	179.6	−72.5	12.9	28
FJ26	8, 5, 2	59.4	65.8	9.3	44
FJ27d	8, 7, 0	40.3	72.4	9.9	38
FJ28	9, 5, 4	46.4	74.0	8.1	41
FJ29	11, 8, 2	74.8	78.6	6.5	55
FJ30	11, 7, 3	34.4	70.5	8.7	32
FJ31	14, 7, 5	47.0	75.1	7.2	38
FJ32	10, 3, 4	66.7	63.3	5.6	118
FJ33d	14, 11, 3	248.1	−59.3	5.9	46
Mean (*g*)	20 sites	48.2	72.3	3.6	85
Mean (*d*)	11 sites	51.3	67.1	5.8	64
Mean (*u*)	31 sites	49.5	70.5	3.0	73
Mean (*t*)	31 sites	77.0	75.0	3.0	73

N, *e*, *r* are the number of specimens measured (*N*) and the number of specimens used in the averages calculated by end-points (*e*) or remagnetization circles (*r*). The mean direction is given by its declination (*D*), inclination (*I*), radius of cone of 95% confidence (α_{95}), and precision parameter (*k*) (Fisher, 1953). Mean (*g*) is for the Fourth of July granitoid sites; mean (*d*) is for the dike sites; mean (*u*) is the overall in situ mean; mean (*t*) is the overall tilt-corrected mean.

[a] d—mafic dikes, s—Surprise Lake Batholith.

coarse-grained magnetite (FJ06, FJ11), and pure or fine-grained magnetite (FJ15, FJ23) (Fig. 6b). SIRM testing shows further that their magnetic carriers comprise multidomain to pseudosingle domain magnetite (FJ04, FJ15) (Fig. 7a) and pseudosingle to single domain pyrrhotite (FJ13) (Fig. 7b). Step demagnetization vector plots show that the dikes have a single stable ChRM component with a minimal VRM overprint that is removed within the first two or three demagnetization steps (Fig. 8c,g).

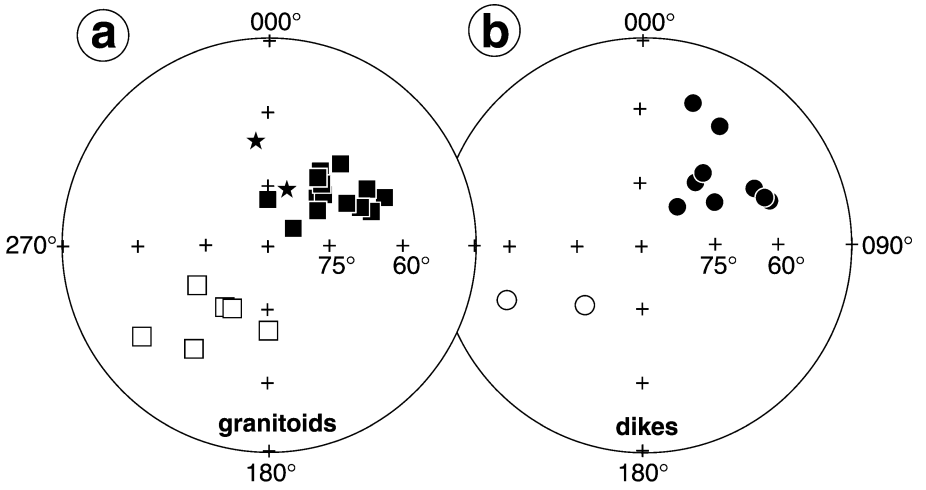
The site mean ChRM directions for the dikes (Table 4) also yielded a bimodal distribution with nine sites directed northeast and steeply down and two sites antiparallel (Fig. 9b). The *F*-statistic of McFadden and Lowes (1981) indicates that the two clusters can be drawn from one population at the 95% confidence level also (i.e., $F_{calc}=0.29 \ll F_{tab}=4.41$). Combining all 11 site mean directions, after reversing the two up-directions, yields an unit mean ChRM direction for the dikes at $D=50.6°$, $I=67.1°$, $\alpha_{95}=5.8°$, and $k=67$ (Table 4).

5.4. Results for the Surprise Lake Batholith

Thermal demagnetization curves for specimens from the Surprise Lake Batholith show magnetite unblocking temperatures (Fig. 6c), although the gradual intensity decay over the entire temperature range suggests that the magnetite is either coarse-grained or titanium-rich. SIRM testing indicates that the dominant magnetic carriers are multidomain to pseudosingle domain magnetite (FJ17) (Fig. 7a). Vector plots for these granites show that they retain a single stable ChRM component with a negligible VRM component (Fig. 8d,h). The mean ChRM direction for the specimens from the two sampling sites is directed to the north–northwest and steeply down (Fig. 9a) at $D=329°$, $I=66°$, $\alpha_{95}=8°$ ($N=15$ specimens) (Table 4).

5.5. Contact tests

At site FJ06 (Fig. 2), specimens from the dike yield a well-defined cluster of ChRM directions to the north and steeply down whereas the host granite (site FJ05) a few metres away has a ChRM direction that is southwest and upward (Table 4). The two granite specimens sampled within 4 cm of the 40-cm-wide dike in its baked zone have NRM directions to the northeast and down, similar to the ChRM direc-



Up sites	Down sites	Up sites	Down sites
N = 6; R = 5.916	N = 14; R = 13.875	N = 2; R = 1.980	N = 9; R = 8.867
α_{95} = 8.75°; k = 60	α_{95} = 3.91°; k = 104	α_{95} = 37°; k = 49	α_{95} = 6.69°; k = 60
D = 218.9°; I = -70.4°	D = 52.8°; I = 73.0°	D = 239.2°; I = -66.5°	D = 48.7°; I = 67.2°
All sites		**All sites**	
N = 20; R = 19.776		N = 11; R = 10.843	
α_{95} = 3.6°; k = 85		α_{95} = 5.8°; k = 64	
D = 48.2°; I = 72.3°		D = 51.3°; I = 67.1°	

Fig. 9. Central part of an equal-area projection showing uncorrected site mean ChRM directions from Table 4. (a) Squares are for the Fourth of July Batholith sites and stars are for the Surprise Lake Batholith sites. (b) Site mean directions for the mafic dikes. Closed (open) symbols represent directions in the lower (upper) hemisphere, indicating normal (reverse) polarity.

tion of the dike (Fig. 10). However, with increasing AF demagnetization steps, their direction trends toward the granite's direction with ChRM directions to the south and shallowly to moderately up. These data constitute a positive contact test and suggest that both the dike and the granite retain a primary remanence.

At site FJ33 (Fig. 2), the dike specimen directions give a fairly well-defined cluster to the southwest and moderately up (Table 4). Conversely, the granite host and a second mafic dike from the same outcrop (sites FJ03 and FJ04) give northeast and down ChRM directions. Two of the three granitoid specimens used in the contact test were collected within 3 cm of the 25 cm-wide dike (FJ33). They have ChRM directions that are similar to those of the dike in the low temperature range but could not be resolved in the high temperature range. A ChRM vector could not be resolved for the third specimen in the baked zone. Although this contact test would have been more conclusive if a high temperature component could have been resolved in the specimens from the baked zone, it does still show that some dikes carry an antiparallel magnetization to their host rocks and this indicates that both the dike and granitoid retain a primary remanence.

The specimens from the two sites in the Surprise Lake Batholith yield a mean ChRM direction to the north and steeply down (Table 4), whereas those in the adjacent granitoids of the older Fourth of July Batholith give a mean ChRM direction that is statistically different at the 95% confidence level. This provides another contact test that indicates that the sampling sites in the Fourth of July Batholith were not regionally remagnetized when the Surprise Lake Batholith was intruded, and hence, that the Fourth

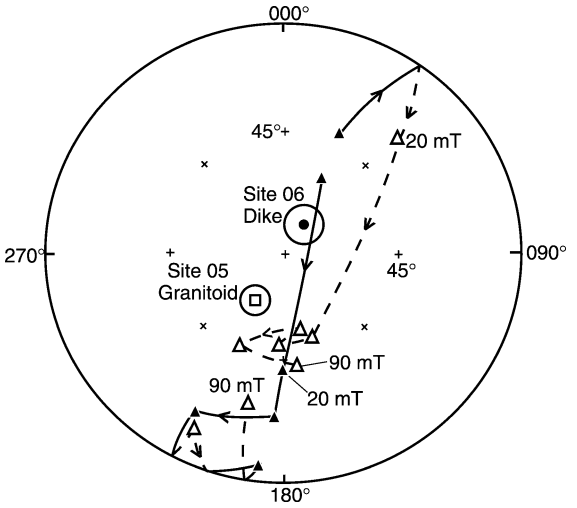


Fig. 10. Equal-area projection showing the results of one contact test. The circle is the site mean for the mafic dike and the square is the site mean for the Fourth of July granitoid a few metres away with their cones of 95% confidence. The triangles are step demagnetization directions of two granitoid specimens collected within the baked zone from the dike (see text). Closed (open) symbols as defined in Fig. 9.

of July Batholith retains a pre-Late Cretaceous magnetization.

6. Discussion

6.1. Age of magnetization

The Fourth of July Batholith is considered to be a post-tectonic intrusion because it shows only minor brittle deformation, has primary flow structures around the margins, and has only minor petrographic alteration (Aitken, 1959; Mihalynuk et al., 1992). Further, most local deformation of the host rocks is thought to have ended by the Early to Middle Jurassic with the imbrication and obduction of the Cache Creek Terrane over the Stikine Terrane (Monger, 1977; Cordey et al., 1991; Mihalynuk et al., 1992).

The similarity of the U–Pb and hornblende $^{40}Ar/^{39}Ar$ ages from the main granodiorite phase at ~ 173 Ma points to relatively rapid initial cooling of the batholith upon intrusion. The equivalence at ~ 165 Ma between the biotite $^{40}Ar/^{39}Ar$ ages from the granodiorite (FJ21) and the eastern lamprophyre dike (FJ01) suggests either: (1) the dike intruded the granodiorite when it was at an elevated temperature and both then cooled through the closure temperature of biotite at ca. 280 °C; or (2) the dike reheated the surrounding granodiorite to >280 °C upon intrusion, thereby resetting the granodiorite biotite systematics to the dike's age and negating any possibility of reconstructing the thermal closure history of the granodiorite. The slightly younger argon age of ~ 162 Ma for the western lamprophyre dike (FJ07) suggests that the dike swarm is slightly younger than the granodiorite or had multiple intrusive events. Further, the thermal effect of these small dike intrusions on the large granodiorite batholith is limited: therefore, scenario (1) above is preferred.

Overall, there is no evidence that a high temperature alteration event has affected the Ar systematics (~ 280 °C), the magnetic properties (>300 °C), or the hornblende chemistry (>500 °C). Therefore, the reported ages, measured pressures, temperatures and ChRM directions are considered primary with the magnetization being acquired at ~ 173 Ma.

6.2. Geobarometric summary

Minor topographic corrections have been made to the geobarometric data. The most eastern sites, i.e., sites FJ24-27, are at the highest elevations, and therefore no correction will be made to them. The sites along the road, just east of Atlin Lake are, on average, 760 m below the most eastern sites. Therefore, 21 MPa will be added to the pressures at sites FJ02, FJ03, FJ29 and FJ31, i.e., 760 m × 0.028 MPa/m = 21 MPa. Sites FJ09 and FJ12 are about 855 m below the most eastern sites, and therefore, 24 MPa will be added to their pressures (Fig. 5).

There is a significant difference in pressures and the resulting crystallization depths in the batholith between the sites on the western shore of Atlin Lake that average 3.6 km and those to the east of the lake that average 9.1 ± 2.2 km. These data suggest differential uplift of ~ 6 km across the lake. Minor fracturing and alteration of granitic outcrops along the shores led Aitken (1959) to first suggest the presence of a fault zone along the lake. Although the apatite fission-track dates are imprecise, they also suggest the presence of a fault, giving similar uplift ages of ~ 65 Ma on the east side of Atlin Lake and an earlier 93 Ma on

the west side (Fig. 2, Table 1) and implying differential uplift on a fault (Donelick and Dickie, 1991). With more concrete geobarometric evidence of substantial differential uplift across Atlin Lake in hand, it is proposed herein to formally name the fault the "Atlin Lake Fault Zone".

The crystallization depths for the two sites on the western shore of Atlin Lake are not significantly different at the 95% confidence level and thus provide no evidence for tilt, i.e., $\pm$ 60 MPa = $\pm$ 2.2 km (Table 3, Fig. 5). East of the lake, the batholith does show a systematic decrease in pressure from the north to the south, implying differential uplift. The differential uplift could record the vertical motion of discrete fault blocks, without tilting of the batholith, however, there are no obvious east–west faults through the batholith to support this hypothesis. Conversely, the batholith may have been tilted as a single entity. If true, then the crystallization depths yield a south–southwest-side down tilt of $\sim 9°$. This possible tilting event is probably not syn-intrusion because the ChRM directions would likely then be "streaked" in an elongated pattern (e.g., Symons et al., 2000a). Therefore, the tilting event is likely post-magnetization and the paleomagnetic directions will require tectonic corrections. Assuming vertical emplacement of the mafic dikes, their present average dip of $\sim 75°$ from horizontal to the north–northeast also implies $\sim 15°$ of south–southwest-side down tilt of the batholith after dike emplacement to support the geobarometric data.

6.3. Tectonic summary

6.3.1. Cache Creek Terrane

The similarity of biotite $^{40}Ar/^{39}Ar$ ages between the lamprophyre dikes and the granodiorite, plus the petrologic similarities, suggest that intrusion of the batholith and the dikes was nearly coeval and, therefore, both likely retain similar ChRM directions. In fact, their cones of 95% confidence overlap both mean directions, which proves that they are statistically the same. The combined in situ unit mean ChRM direction is at $D = 49.5°$, $I = 70.5°$, $\alpha_{95} = 3.0°$, and $k = 73$ ($N = 31$ sites; Table 4), which yields a 173-Ma paleopole at 54.9°W, 63.4°N ($d_p = 4.5°$, $d_m = 5.2°$). Compared to the 173-Ma reference pole of 249°W, 67.5°N ($\alpha_{95} = 3.5°$) that is interpolated from the apparent polar wander path (APWP) of Besse and Courtillot (1991), the unit paleopole is near-sided, suggesting that the batholith and hosting Cache Creek Terrane have been translated away from the reference pole or "southward" by $9.0 \pm 3.7°$ (1000 ± 410 km) and rotated clockwise by $79 \pm 6°$ (Fig. 11). Correcting for the 9° south–southwest-side down tilt towards $\sim 190°$ yields a unit mean ChRM direction of $D = 077°$, $I = 75°$, $\alpha_{95} = 3.0°$, and $k = 73$ (Table 4) and an associated paleopole of 81.2°W, 54.6°N ($d_p = 5.0°$, $d_m = 5.5°$). Comparison of this paleopole with the 173-Ma reference pole suggests that the batholith and hosting Cache Creek Terrane have been significantly translated away from the reference pole or "southward" by $16.1 \pm 3.7°$ (1790 ± 410 km) and rotated clockwise by $107 \pm 7°$ (Fig. 11).

Three other Jurassic reference poles for the North American craton (e.g., Gordon et al., 1984; May and Butler, 1986; Van der Voo, 1993) have been compared with the tectonic-corrected pole from the Fourth of July Batholith (Fig. 11). Use of one of these other reference poles would increase the southward translation estimate by 3° to 4° and the rotation estimate by $\pm 4°$.

The tectonically corrected tectonic motion estimate of the Fourth of July Batholith is dissimilar to that previously published for the Atlin area supracrustal rocks. Cole et al. (1992) suggested that the host rocks for the Fourth of July Batholith showed no translation and only minor counter-clockwise rotation assuming Early Jurassic remagnetization. However, their results are considered with caution because there is no age control on either the remagnetization or deformation events. Further, their two sampling areas are small and may not average out secular variation, particularly if the ChRM is a remagnetization component as suggested by the fact that all of their samples retain a normal polarity despite being supposedly acquired during a time period with many polarity changes (e.g., Cox, 1982; Opdyke and Channell, 1996), as is found for the Fourth of July Batholith and associated dikes. Also, their two sampling sites are located on either side of the then unrecognized Atlin Lake Fault Zone, so that the tilt on the eastern side of the lake was not in consideration, which could affect the data of their southern sampling site.

6.3.2. Stikine Terrane

The 206–190-Ma Hazelton–Takla volcanic rocks that are exposed in the Stikine Terrane in central

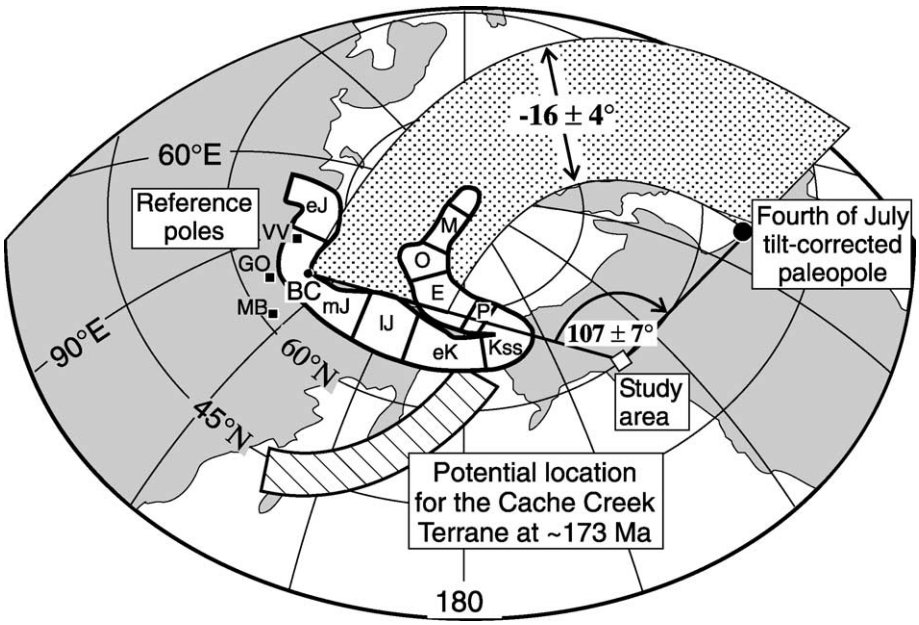


Fig. 11. Locations of the Fourth of July study area, its tilt-corrected paleopole, the interpolated 173 Ma reference pole and APWP of Besse and Courtillot (1991; BC). Other Jurassic North American craton reference poles are from: VV—Van der Voo (1993); GO—Gordon et al., 1984; MB—May and Butler (1986).

British Columbia (Fig. 1) also show dissimilar estimates of translation but large rotations. Monger and Irving (1980) obtained an estimate of ~ 1300 km of northward translation and clockwise rotations of about − 103° to 14° for three small sampling areas. Using two different reference poles, Gordon et al. (1984) and May and Butler (1986) suggested that there was no translation. Vandall and Palmer (1990) obtained a translation estimate of 0 ± 10° and clockwise rotation estimates of 14°, 79° and 117° for three different areas of these volcanic rocks. The variability in the rotation values is not surprising because the sampled units dip steeply and thus likely incorporate plunge estimate errors, and because their primary remanence is carried by both magnetite and hematite and thus likely incorporates errors from the incomplete demagnetization of the specimens (e.g., Monger and Irving, 1980). However, the mean clockwise rotation found for the Hazelton–Takla Group of 58 ± 47° is similar to clockwise rotation estimates found for Cretaceous rock units in the Intermontane Belt, supporting the validity of the Hazelton–Takla Group tectonic estimates.

Paleomagnetic results for the ~ 177-Ma Teslin Crossing Stock in the Stikine terrane of the southern Yukon yielded a southward translation estimate of 21.1 ± 7.4° and a clockwise rotation of 33 ± 18°, thus having translation values that are similar to the results for the Fourth of July Batholith (Table 5). There is no evidence for tilt of the Teslin Crossing Stock; biotite geothermometry does not statistically support a tilt, nor do the horizontal host rocks (Harris et al., 1999a).

6.3.3. Quesnel Terrane

Two studies in southern British Columbia provide Jurassic tectonic estimates for the Quesnel Terrane. The ~ 198 Ma Guichon Batholith yielded a revised geobarometrically corrected paleomagnetic result that implies a 13.8 ± 6.6° northward translation and 43 ± 7° clockwise rotation for Quesnellia (Harris, 1998; Harris et al., 1998). These values correct for the measured 6.0° tilt down to the northeast, but they are not statistically significantly different from the original values (Symons, 1971, 1983). The ~ 203-Ma Copper Mountain Intrusions could not be tested geo-

Table 5
Summary of paleopoles and tectonic estimates for the Intermontane Belt

Collection	Age (Ma)	Paleopole (°E, °N)	d_p, d_m (°)	Translation (+°N)	Rotation (+°CW)
Carmacks Volcanics[a]	70	88.6, 78.4	7.1, 8.4	20.6 ± 7.0	20 ± 10
Mount Lorne Stock[b]	75	290.9, 78.3	4.1, 4.5	10.5 ± 3.5	57 ± 11
Axelgold Intrusions[c]	103	327.0, 76.0	7.0, 8.0	13.9 ± 6.0	55 ± 13
Spences Bridge Volc.[d]	104	321.0, 64.4	6.8, 8.6	9.5 ± 5.7	66 ± 13
Mount McIntyre Stock[e]	109	299.6, 70.5	5.5, 6.1	14.3 ± 5.0	79 ± 16
Whitehorse Batholith[f]	112	285.5, 81.7	5.3, 5.7	11.0 ± 4.8	59 ± 17
Endako Intrusions[g,h]	142	121.8, 58.4	8.5, 12.1	20.5 ± 10.0	2 ± 15
Fourth of July—in situ	172	305.1, 63.4	4.5, 5.2	−9.0 ± 3.7	79 ± 6
tilt-corrected		278.8, 56.6	5.0, 5.5	−16.1 ± 3.7	107 ± 7
Teslin Crossing Stock[i]	177	249.7, 84.3	15.5, 16.5	−21.0 ± 7.4	33 ± 18
Guichon Int.[g,j]—in situ	198	12.2, 52.4	5.0, 8.5	11.0 ± 6.7	43 ± 7
tilt-corrected		13.5, 49.6	4.6, 8.2	13.8 ± 6.6	43 ± 7
Copper Mountain Int.[k]	203	11.8, 57.3	2.7, 4.4	9.0 ± 5.0	39 ± 6
Hazelton-Takla Volc.[l]	206				

CW, clockwise.
[a] Wynne et al. (1998).
[b] Harris et al. (1999b).
[c] Monger and Irving (1980).
[d] Irving and Thorkelson (1990), Irving et al. (1996).
[e] Harris et al. (1996), only the northeast mean direction is used.
[f] Harris et al. (1997).
[g] Harris (1998).
[h] Symons (1973).
[i] Harris et al. (1999a).
[j] Symons (1971, 1983).
[k] Symons and Litalien (1984).
[l] Vandall and Palmer (1990) and Monger and Irving (1980), see text.

barometrically for tilt because of the lack of hornblende, but their original tectonic estimates similarly suggested a northward translation of 9.0 ± 5.0° and clockwise rotation of 39 ± 6° (Symons and Litalien, 1984). The distinct differences in their Jurassic paleomagnetic data with respect to the previously mentioned Jurassic data suggest that Quesnellia did not amalgamate with the Cache Creek and Stikine terranes until the Late Jurassic (Harris, 1998).

6.3.4. Yukon-Tanana Terrane

The ~ 190 Ma Big Creek Batholith lies within the Yukon-Tanana Terrane in central Yukon (Fig. 1). Using the 190-Ma reference pole, the geobarometrically corrected paleomagnetic results suggest that this terrane is 9.3 ± 6.1° near-sided and rotated counter-clockwise by 32 ± 11°. Although the translation estimate is relatively similar to the Fourth of July estimate, the counter-clockwise rotation is opposite in sense to that found in Stikine and Cache Creek rock units, but has recently been found elsewhere in the Yukon-Tanana Terrane (e.g., McCausland et al., 1998; Symons et al., 2000b). Thus, the Yukon-Tanana Terrane appears to have been a parautochthon to the North American craton since the Early Jurassic, and to have had a very different tectonic history than the Stikine and Cache Creek terranes (Symons et al., 2000b).

6.3.5. Intermontane assembly

These southern translation estimates from Jurassic rocks in the Cache Creek and Stikine terranes contrast with the northward estimates from Cretaceous and Paleocene units in these same terranes (Table 5). This implies a dramatic change in the tectonic evolution of the Intermontane Belt terranes during the Late Jurassic. Specifically, the translation and rotation estimates for the Fourth of July Batholith indicate that the Cache Creek Terrane was located at a latitude north of the North American craton during the Early Jurassic, and likely outboard of the North American Craton within

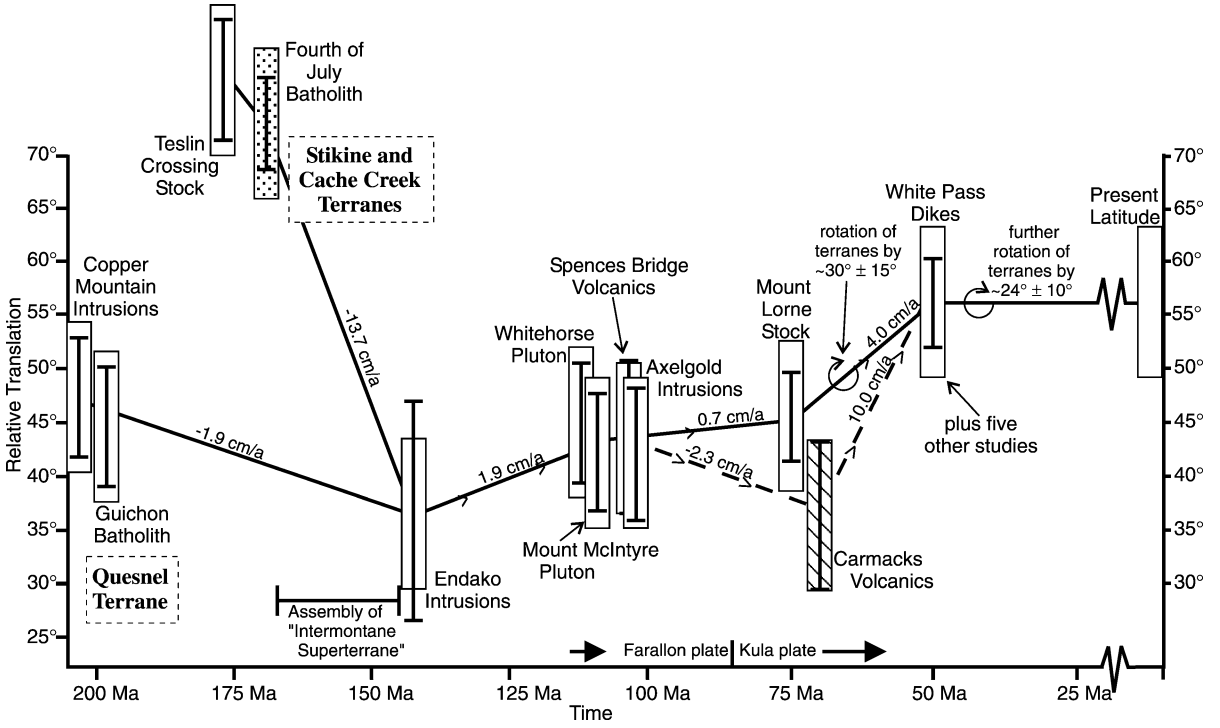


Fig. 12. Plot of relative translation distances for the Intermontane Belt versus time (after Harris, 1998). Blocks represent scaled dimensions of the present-day Intermontane Belt. Error bars on the blocks represent the 95% confidence errors on the translation estimates (Table 5). Farallon and Kula plates at the bottom refer to the oceanic plate associated with the tectonic motion from Engebretson et al. (1985).

the ancestral Pacific Basin (Fig. 12). Thus, the results support plate tectonic models of the Pacific Ocean Basin that suggest that the Farallon oceanic plate had a dominantly southeasterly motion prior to the Cretaceous (Engebretson et al., 1985; Kelley, 1993; Van der Voo, 1993). This southeasterly motion would transport the Stikine and Cache Creek terranes across the Pacific Basin to the North American Craton. By the middle of the Early Cretaceous, the amalgamated terranes of the Intermontane Belt had begun northward translation along the cratonic margin (Fig. 12; Irving et al., 1996; Harris et al., 1997, 1999b; Harris, 1998).

7. Conclusions

Argon geochronology results on hornblende separates indicate that the Fourth of July Batholith is ~ 173 Ma old, in agreement with U–Pb zircon analyses. Further, dates for biotite separates indicate that the batholith and associated mafic dikes cooled through ~ 280 °C at ~ 165 Ma. Geobarometric analyses indicate the presence of a north–south fault running along the length of Atlin Lake with the east side having been uplifted by ~ 6 km more relative to the west side. Further, the Fourth of July Batholith appears to have been tilted uniformly down by ~ 9° towards the south–southwest at ~ 190°. Tilt-corrected paleomagnetic analyses suggest that the Cache Creek Terrane has been significantly translated $16 \pm 4°$ southward and rotated $107 \pm 7°$ clockwise since the Middle Jurassic. This result is in agreement with the Jurassic Teslin Crossing Stock translation estimate from the Stikine Terrane, but not with other estimates from the terranes underlying the Intermontane Belt. Further, the proposed tectonic model suggests that the Stikine and Cache Creek terranes were at a similar paleolatitude north of the North American craton at ~ 175 Ma and far off the margin, before moving southeasterly towards the craton to amalgamate with Quesnellia by the mid to Late Jurassic. This

model is supported by the oceanic plate motion models of Engebretson et al. (1985) and Van der Voo (1993).

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